

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDEEN
YEAR/SEM/SEC: Y2/S4/B

FIBONACCI SERIES

SOURCE CODE:

```
package fibonacci;

import java.util.Scanner;

public class Fibonacci {

    public static void main(String[] args) {

        int count,num1=0,num2=1,sum2=1;

        System.out.println("how many numbers you want in the sequence");

        Scanner Sc=new Scanner(System.in);

        count=Sc.nextInt();

        Sc.close();

        System.out.println("fibonacci series of "+count+"number=");

        int i=1;

        while(i<=count)

        {

            System.out.println(num1+"");

            int SumofprevTwo =num1+num2;

            num1 =num2;

            num2 =SumofprevTwo;

            i++;

        }

    }

}
```

OUTPUT:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CCL2\Desktop\fibonacciseries\src\fibonacciseries>javac Fibonacci1.java

C:\Users\CCL2\Desktop\fibonacciseries\src\fibonacciseries>java Fibonacci1
how many numbers you want in the sequence
10
fibonacci series of 10 numbers=
0
1
1
2
3
5
8
13
21
34

C:\Users\CCL2\Desktop\fibonacciseries\src\fibonacciseries>_
```

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YEAR/SEM/SEC: Y2/S4/B

SWITCH CASE

SOURCE CODE:

```
package switchcase;

import java.util.Scanner;

public class Switchcase {

    public static void main(String args[]) {

        char c;

        int firstnumber,secondnumber,output;

        Scanner Sc=new Scanner (System.in);

        System.out.println("Enter +for addition,-for subtraction,*for multiplication,/for
division:");

        c=Sc.next().charAt(0);

        System.out.println("enter frist number:");

        firstnumber=Sc.nextInt();

        System.out.println("Enter the second number;");

        secondnumber=Sc.nextInt();

        switch(c){

            case '+':

                output=firstnumber+secondnumber;

            case '-':

                output=firstnumber-secondnumber;

                break;

            case '*':

                output=firstnumber*secondnumber;
```

```
        break;
    case '/':
        output=firstnumber/secondnumber;
        break;
    default:
        System.out.println("please emter the valid symbol!!!");
        return;
    }
    System.out.printf("%d%s%d=%d",firstnumber,c,secondnumber,output);
}
}
```

OUTPUT:

```
C:\Windows\System32\cmd.exe

C:\Users\CCL2\Desktop\Switchcase>javac Switch.java

C:\Users\CCL2\Desktop\Switchcase>java Switch
Enter +for addition,-for subtraction,*for multiplication,/for division:
+
enter frist number:
8
Enter the second number;
5
8+5=3
C:\Users\CCL2\Desktop\Switchcase>java Switch
Enter +for addition,-for subtraction,*for multiplication,/for division:
-
enter frist number:
6
Enter the second number;
2
6-2=4
C:\Users\CCL2\Desktop\Switchcase>java Switch
Enter +for addition,-for subtraction,*for multiplication,/for division:
*
enter frist number:
8
Enter the second number;
7
8*7=56
C:\Users\CCL2\Desktop\Switchcase>java Switch
Enter +for addition,-for subtraction,*for multiplication,/for division:
/
enter frist number:
84
Enter the second number;
6
84/6=14
C:\Users\CCL2\Desktop\Switchcase>
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDEEN
YEAR/SEM/SEC: Y2/S4/B

CLASSES AND OBJECTS

SOURCE CODE:

```
package Classandobjects;

import java.util.Scanner;

public class Classandobjects
{
    String name;

    float min_bal,amt,w;

    long account_no;

    void assign()
    {
min_bal=1000;
    }

    void input()
    {
        Scanner no=new Scanner(System.in);

System.out.println("Enter user name: ");

        name=no.nextLine();

        Scanner sr=new Scanner(System.in);

System.out.println("Enter the account number");

account_no=sr.nextLong();

System.out.println("Enter the amount you want to deposit: ");

amt=sr.nextFloat();
```

```

        min_bal=min_bal+amt;
    }

    void withdraw()
    {
        System.out.println("Enter the amount you want to withdraw: ");

        Scanner s1=new Scanner(System.in);

        w=s1.nextFloat();
        min_bal=min_bal-w;
    }

    void display()
    {
        System.out.println("-----INDIAN BANK-----");
        System.out.println("Account holder name: "+name);
        System.out.println("Account number: "+account_no);
        System.out.println("Account amount: "+min_bal);
    }

    public static void main(String[] args){
        Classandobjects b=new Classandobjects();
        b.assign();
        b.input();
        b.withdraw();
        b.display();
    }
}

```

OUTPUT:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CCL2\Desktop\Classes and Objects>javac Classandobjects.java

C:\Users\CCL2\Desktop\Classes and Objects>java Classandobjects
Enter user name:
Tony Stark
Enter the account number
4562153245
Enter the amount you want to deposit:
5000
Enter the amount you want to withdraw:
600
-----INDIAN BANK-----
Account holder name: Tony Stark
Account number: 4562153245
Account amount: 5400.0

C:\Users\CCL2\Desktop\Classes and Objects>
```


REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

CONSTRUCTOR AND DESTRUCTOR

SOURCE CODE:

```
package constructor;  
  
import java.util.*;  
  
public class Studentdetails {  
    public static void main(String[] args)  
    {  
        Student stud = new Student();  
stud.input();  
stud.calc();  
stud.display();  
stud.finalize();  
    }  
}  
  
class Student  
{  
    int s1,s2,s3,s4,s5,i,avg,tot;  
    String name;  
    Scanner s = new Scanner(System.in);  
Student()  
    {  
        System.out.println("Student Details");  
        System.out.println("*****");  
    }  
}
```

```
}  
  
public void finalize()  
{  
    System.out.println("Verified");  
}  
  
void input()  
{  
    System.out.println("Enter your name:");  
    name = s.nextLine();  
    System.out.println("Enter the marks: ");  
    s1 = s.nextInt();  
    s2 = s.nextInt();  
    s3 = s.nextInt();  
    s4 = s.nextInt();  
    s5 = s.nextInt();  
}  
  
void calc()  
{  
    tot = s1+s2+s3+s4+s5;  
    avg = tot/5;  
}
```

OUTPUT:



The screenshot shows a debugger window with multiple tabs. The active tab is 'Calculator (debug)'. The output pane displays the following text:

```
debug:
student details
*****
Enter your name:
Dawid
enter the marks
90
98
95
96
93
name=Dawid
marek=90
98
95
96
93
total=472
average=94

Grade=S
verified
BUILD SUCCESSFUL (total time: 12 seconds)
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

METHOD OVERLOADING

SOURCE CODE:

```
package methodoverloading;

import java.io.*;
import java.lang.*;
import java.lang.Math;
import java.util.*;
import java.io.IOException;

public class methodoverloading {

    public static void main(String[] args) throws IOException
    {
        shape s = new shape();

        int l,b;

        int a;

        String ss;

        BufferedReader brr = new BufferedReader (new InputStreamReader(System.in));

        System.out.println("Enter the length (rectangle):");

        ss = brr.readLine();

        l = Integer.parseInt(ss);

        System.out.println("Enter the breadth (rectangle):");

        ss = brr.readLine();

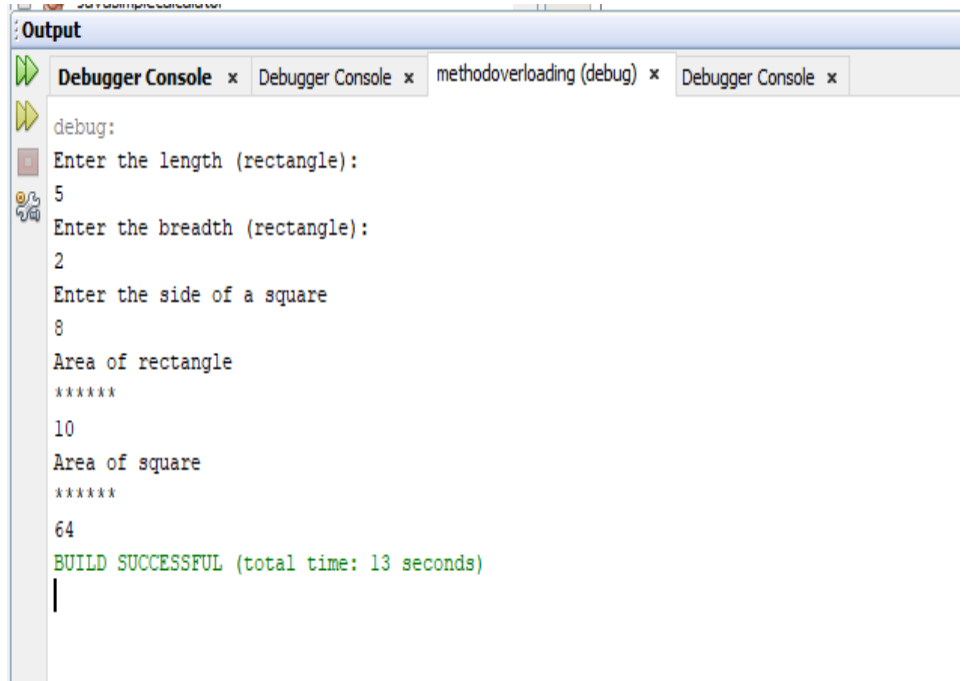
        b = Integer.parseInt(ss);

        System.out.println("Enter the side of a square");

        ss = brr.readLine();
```

```
        a = Integer.parseInt(ss);
s.area(l,b);
s.area(a);
    }
}
class shape
{
    void area(int l, int b)
    {
        System.out.println("Area of rectangle");
        System.out.println("*****");
        int c = l*b;
        System.out.println(c);
    }
    void area(int a)
    {
        System.out.println("Area of square");
        System.out.println("*****");
        int d = a*a;
        System.out.println(d);
    }
}
```

OUTPUT:



The screenshot shows the 'Output' window of an IDE. It contains several tabs: 'Debugger Console', 'Debugger Console', 'methodoverloading (debug)', and 'Debugger Console'. The active tab is 'methodoverloading (debug)'. The output text is as follows:

```
debug:  
Enter the length (rectangle):  
5  
Enter the breadth (rectangle):  
2  
Enter the side of a square  
8  
Area of rectangle  
*****  
10  
Area of square  
*****  
64  
BUILD SUCCESSFUL (total time: 13 seconds)  
|
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

MULTIPLE INHERITANCE

SOURCE CODE:

```
package multipleinheritance;

interface MotorBike
{
    int speed=50;
    public void totalDistance();
}

interface Cycle
{
    int distance=150;
    public void speed();
}

public class Multipleinheritance implements MotorBike,Cycle
{
    int totalDistance;
    int avgSpeed;
    public void totalDistance()
    {
        totalDistance=speed*distance;
        System.out.println("Total Distance Travelled : "+totalDistance);
    }
    public void speed()
    {
```

```
int avgSpeed=totalDistance/speed;

System.out.println("Average Speed maintained : "+avgSpeed);

}

public static void main(String args[])

{

Multipleinheritance t1=new Multipleinheritance();

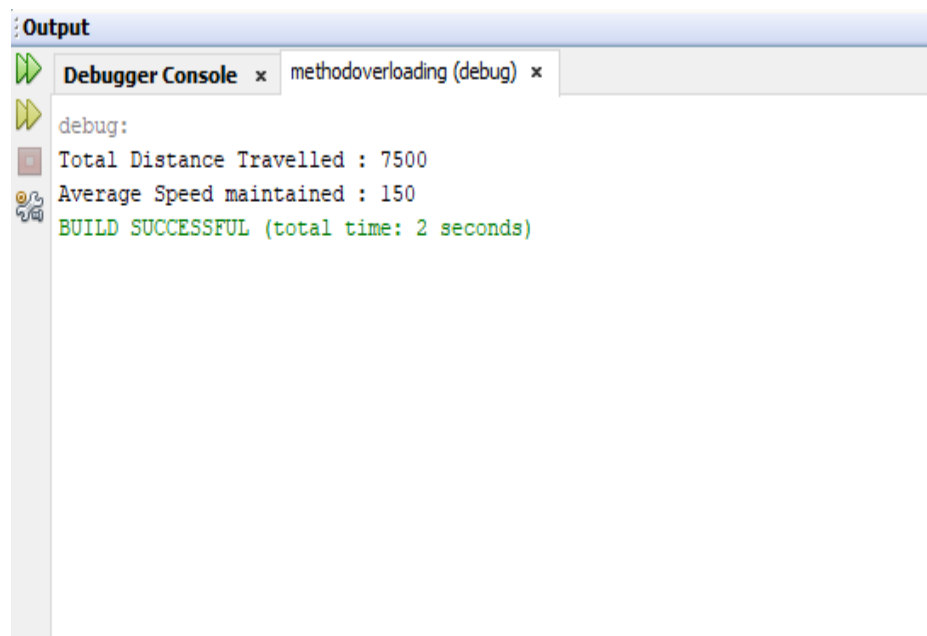
t1.totalDistance();

t1.speed();

}

}
```


OUTPUT:



REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

HIERARCHIAL INHERITANCE

SOURCE CODE:

```
package inheritance;

class product
{
    String name;
    double price;
    public product(String name,double price)
    {
        this.name=name;
    }
    this.price=price;
    }
    public double Calculatetotalprice(int quantity)
    {
        return price*quantity;
    }
}

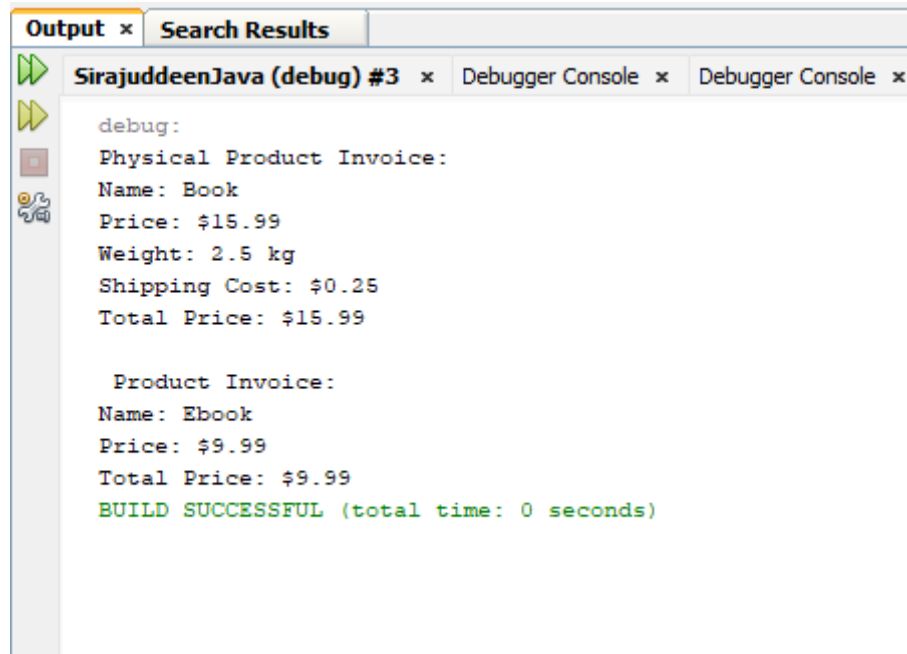
class PhysicalProduct extends product
{
    public double weight;
    public PhysicalProduct(String name,doubleprice,double weight)
    {
        super(name,price);
    }
}
```

```

this.weight=weight;
}
public double CalculateShippingCost()
{
return weight*0.1;
}}
class DigitalProduct extends product
{
public DigitalProduct(String name,double price)
{
super(name,price);
}}
public class hierarchical
{
    public static void main(String args[])
    {
PhysicalProductphysicalProduct=new PhysicalProduct("book",15.99,2.5);
DigitalProductdigitalProduct=new DigitalProduct("Ebook",9.99);
System.out.println("Physical Product Invoice:");
System.out.println("Name:"+physicalProduct.name)
System.out.println("Price:"+physicalProduct.price);
System.out.println("Weight:"+physicalProduct.weight+"kg");
System.out.println("Shipping cost:"+physicalProduct.CalculateShippingCost());
System.out.println("Total price:"+physicalProduct.Calculatetotalprice(1));
System.out.println("\n Product Invoice");
System.out.println("Name:"+digitalProduct.name);
System.out.println("Price:"+digitalProduct.price);
System.out.println("Total price:"+digitalProduct.Calculatetotalprice(1));} }

```

OUTPUT:



```
debug:
Physical Product Invoice:
Name: Book
Price: $15.99
Weight: 2.5 kg
Shipping Cost: $0.25
Total Price: $15.99

Product Invoice:
Name: Ebook
Price: $9.99
Total Price: $9.99
BUILD SUCCESSFUL (total time: 0 seconds)
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

EXCEPTION HANDLING

SOURCE CODE:

```
package exception;

import java.io.*;

import java.io.IOException;

public class Exception{

    public static void main(String[] args)throws IOException

    {

        int ch;

        BufferedReader br=new BufferedReader

            (new InputStreamReader(System.in));

        System.out.println("1.arithmeticException 2.NumberFormatException

3.ArrayIndexOutOfBoundsException 4.StringIndexOutOfBoundsException");

        System.out.println("Enter your choice");

        ch=Integer.parseInt(br.readLine());

        switch(ch)

        {

            case 1:

try{

                int a=30,b=0;

                int c=a/b;

                System.out.println("Result="+c);

            }

}
```

```

catch(ArithmeticException e)
    {
System.out.println("Exception Handling");
System.out.println("-----");
System.out.println("can't divide a number by 0");
    }
    break;
case 2:
try{
        int num=Integer.parseInt("akki");
System.out.println(num);
    }
catch(NumberFormatException e)
    {
System.out.println("Exception Handling");
System.out.println("-----");
System.out.println("Number format exception");
    }
    break;
case 3:
try{
        int a[]=new int[5];
a[6]=9;
    }
catch(ArrayIndexOutOfBoundsException e)
    {
System.out.println("exception Hnadling");
System.out.println("-----");

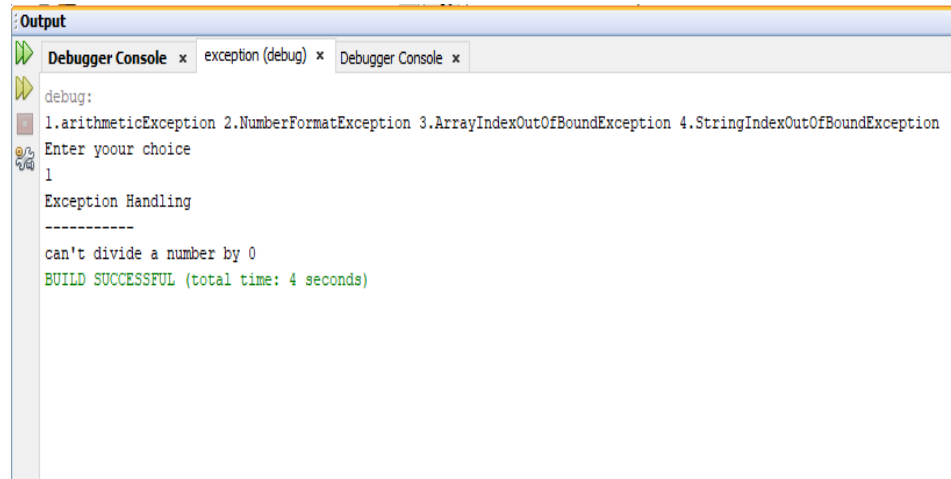
```

```
System.out.println("Array Index is Out of bounds");
    }
    break;
case 4:
    try
    {
        String a="this is like chipping";
        char c=a.charAt(24);
System.out.println(c);
    }
catch(StringIndexOutOfBoundsException e)
    {
System.out.println("exception handling");
System.out.println("-----");
System.out.println("StringIndexOutOfBoundsException");
    }
    break;
default:
System.out.println("INVALID");
    break;
}

}

}
```

OUTPUT:



```
Output
Debugger Console x exception (debug) x Debugger Console x
debug:
1.arithmeticException 2.NumberFormatException 3.ArrayIndexOutOfBoundsException 4.StringIndexOutOfBoundsException
Enter your choice
1
Exception Handling
-----
can't divide a number by 0
BUILD SUCCESSFUL (total time: 4 seconds)
```


REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

METHOD OVERLOADING

SOURCE CODE:

```
package methodoverloading;

import java.io.*;
import java.lang.*;
import java.lang.Math;
import java.util.*;
import java.io.IOException;

public class overloading
{
    public static void main(String[] args)throws IOException
    {
        shape s=new shape();
        int l,b;
        int a;
        String ss;
        BufferedReader brr=new BufferedReader(new InputStreamReader(System.in));
        System.out.println("enter the value");
        ss=brr.readLine();
        l=Integer.parseInt(ss);
        System.out.println("enter the value");
        ss=brr.readLine();
        b=Integer.parseInt(ss);
        System.out.println("enter a value");
```

```

        ss=brr.readLine();

        a=Integer.parseInt(ss);

        s.area(l,b);

        s.area(a);
    }

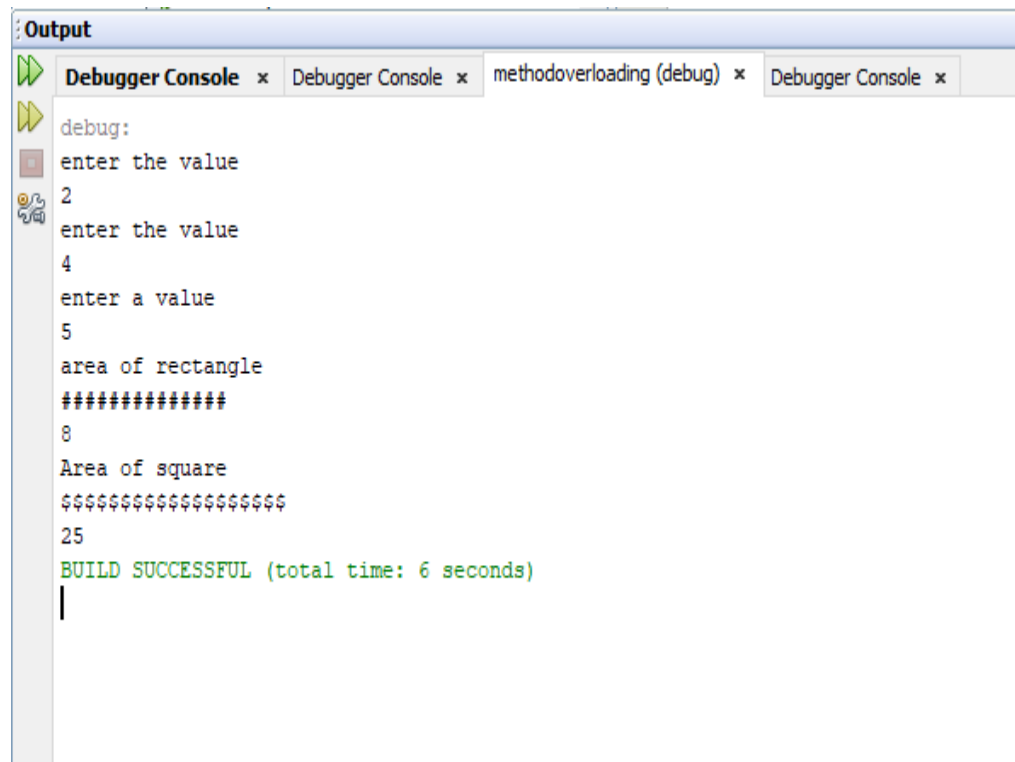
}

class shape
{
    void area(int l,int b)
    {
        System.out.println("area of rectangle");
        System.out.println("#####");
        int c=l*b;
        System.out.println(c);
    }

    void area(int a)
    {
        System.out.println("Area of square");
        System.out.println("$$$$$$$$$$$$$$$$");
        int d= a*a;
        System.out.println(d);
    }
}

```

OUTPUT:



The screenshot shows an IDE's Output window with a tab titled 'methodoverloading (debug)'. The output text is as follows:

```
debug:
enter the value
2
enter the value
4
enter a value
5
area of rectangle
#####
8
Area of square
$$$$$$$$$$$$$$$$$$$
25
BUILD SUCCESSFUL (total time: 6 seconds)
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

FILES AND STREAMS

SOURCE CODE:

```
package filesandstream;

import java.io.File;
import java.io.FileWriter;

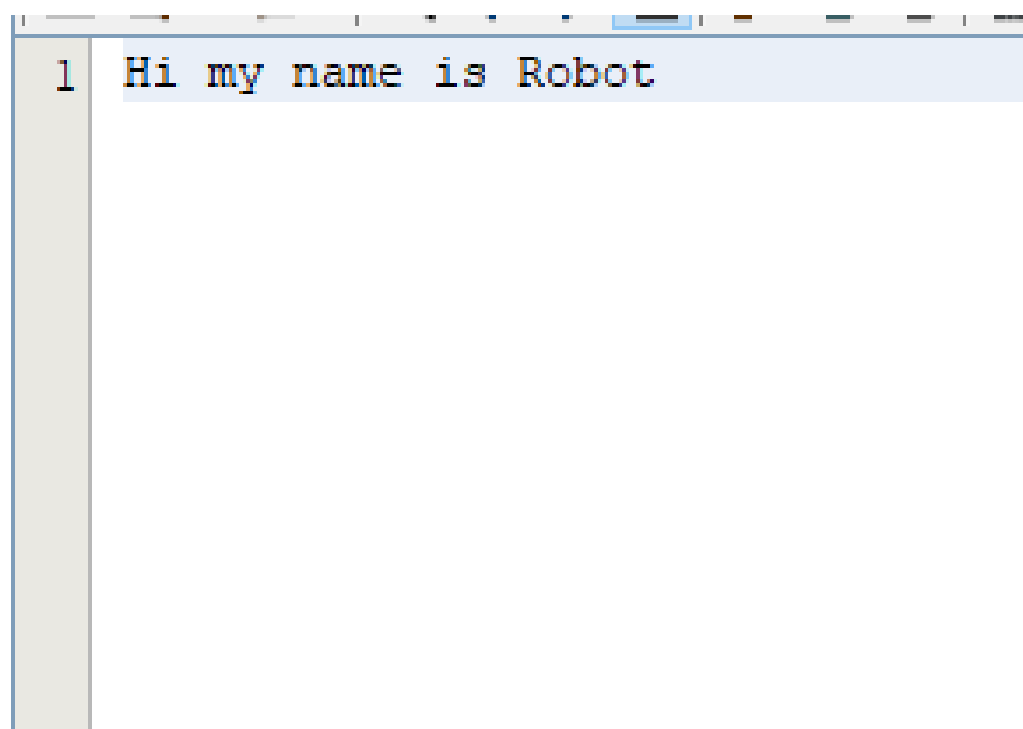
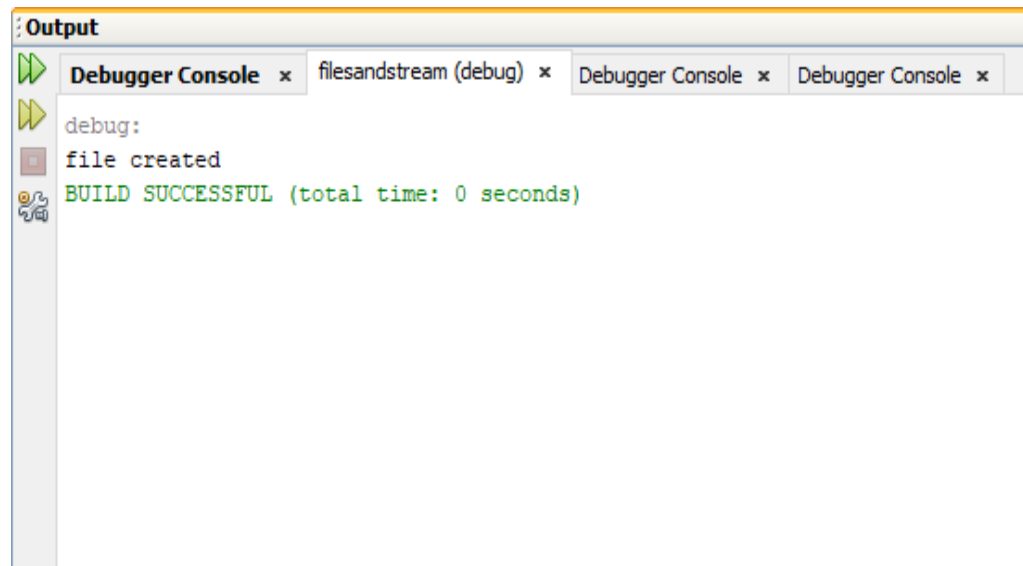
import java.io.IOException;

public class Filesandstream
{
    public static void main(String[] args)
    {
        File file = new File("intro.txt");
        boolean result;

        try
        {
            result = file.createNewFile();
            if(result)
            {
                System.out.println("file created");
            }
            else
            {
                System.out.println("File already exists");
            }
        }
    }
}
```

```
FileWriter myWriter = new FileWriter("intro.txt");  
myWriter.write("Hi my name is Robot");  
myWriter.close();  
}  
catch (IOException e)  
{  
e.printStackTrace();  
}  
}  
}
```

OUTPUT:



REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

PACKAGES

SOURCE CODE:

Mainpack.java:

```
package mypack;

import java.io.*;

import java.io.IOException;

public class MainPack {

    public static void main(String args[]) throws IOException
    {
        Package1 a = new Package1();
        Package2 b = new Package2();
        Package3 c = new Package3();
        DataInputStream obj = new DataInputStream(System.in);
        while(true)
        {
            System.out.println("Choose an option");
            System.out.println("\n\t1. Sum of N numbers");
            System.out.println("\n\t2. Print Odd Number");
            System.out.println("\n\t3. Print Even Number");

            String choice1 = obj.readLine();
            int choice = Integer.parseInt(choice1);
```

```

switch(choice)
{
    case 1:
        a.sumofNnumbers();
        break;
    case 2:
        b.oddNumbers();
        break;
    case 3:
        c.evenNumbers();
        break;
    case 4:
        System.exit(1);
        break;
}
}
}
}

```

Package1.java:

```
// SUM OF N NUMBERS
```

```

package mypack;
import java.io.*;
import java.util.Scanner;
public class Package1 {
    public void sumofNnumbers()
    {
        int n, i=0, s=0;

```



```

        System.out.println("Enter value: ");

        Scanner sr = new Scanner(System.in);

        n = sr.nextInt();

        while(i<n)

        {

            i=i+1;

            System.out.println(i);

            s = s+i;

        }

        System.out.println(s);

    }

}

```

Package2.java:

```

// ODD NUMBER

package mypack;

import java.io.*;

import java.util.Scanner;

public class Package2 {

    public void oddNumbers()

    {

        int n, i=1;

        System.out.println("Enter n value: ");

        Scanner sr = new Scanner(System.in);

        n = sr.nextInt();

        while(i<n)

        {

            i = i+2;

```

```
        System.out.println(i);
    }
}
}
```

Package3.java:

```
// EVEN NUMBER

package mypack;

import java.io.*;
import java.util.Scanner;

public class Package3 {
    public void evenNumbers()
    {
        int n, i=0;

        System.out.println("Enter n value: ");
        Scanner sr = new Scanner(System.in);
        n = sr.nextInt();
        while(i<n)
        {
            i = i+2;

            System.out.println(i);
        }
    }
}
```

OUTPUT:

```
Output
Debugger Console x Debugger Console x Debugger Console x JavaApplication28 (debug) x Debugger Console x
debug:
Choose an option
1. Sum of N numbers
2. Print Odd Number
3. Print Even Number
1
Enter value:
6
1
2
3
4
5
6
21
Choose an option
1. Sum of N numbers
2. Print Odd Number
3. Print Even Number
3
Enter n value:
5
2
4
6
Choose an option
1. Sum of N numbers
2. Print Odd Number
3. Print Even Number
|
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

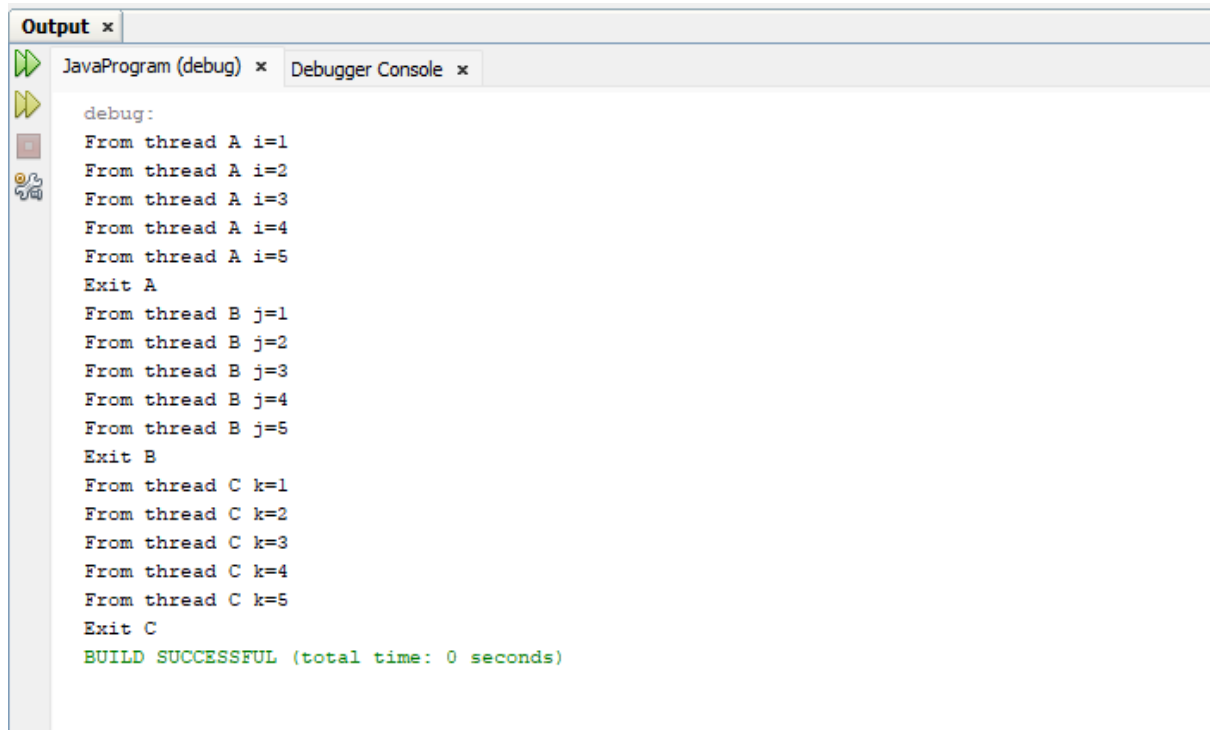
MULTITHREADING

SOURCE CODE:

```
package Javaprogram;
import java.lang.Thread;
class A extends Thread
{
    public void run()
    {
        for (int i=1;i<=5;i++)
        {
            System.out.println("From thread A i="+i);
        }
        System.out.println("Exit A");;
    }
}
class B extends Thread
{
    public void run()
    {
        for(int j=1;j<=5;j++)
        {
            System.out.println("From thread B j="+j);
        }
        System.out.println("Exit B");
```

```
    }  
}  
class C extends Thread  
{  
    public void run()  
    {  
        for(int k=1;k<=5;k++)  
        {  
            System.out.println("From thread C k="+k);  
        }  
        System.out.println("Exit C");  
    }  
}  
class Threadtest  
{  
    public static void main(String[] args)  
    {  
        new A().start();  
        new B().start();  
        new C().start();  
    }  
}
```

OUTPUT:



The screenshot shows an IDE's Output window with two tabs: "Output x" and "Debugger Console x". The "Debugger Console x" tab is active, displaying the following debug output:

```
debug:
From thread A i=1
From thread A i=2
From thread A i=3
From thread A i=4
From thread A i=5
Exit A
From thread B j=1
From thread B j=2
From thread B j=3
From thread B j=4
From thread B j=5
Exit B
From thread C k=1
From thread C k=2
From thread C k=3
From thread C k=4
From thread C k=5
Exit C
BUILD SUCCESSFUL (total time: 0 seconds)
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

SOCKET

SOURCE CODE:

serverside.java:

```
package JavaApplication24;

import java.net.*;

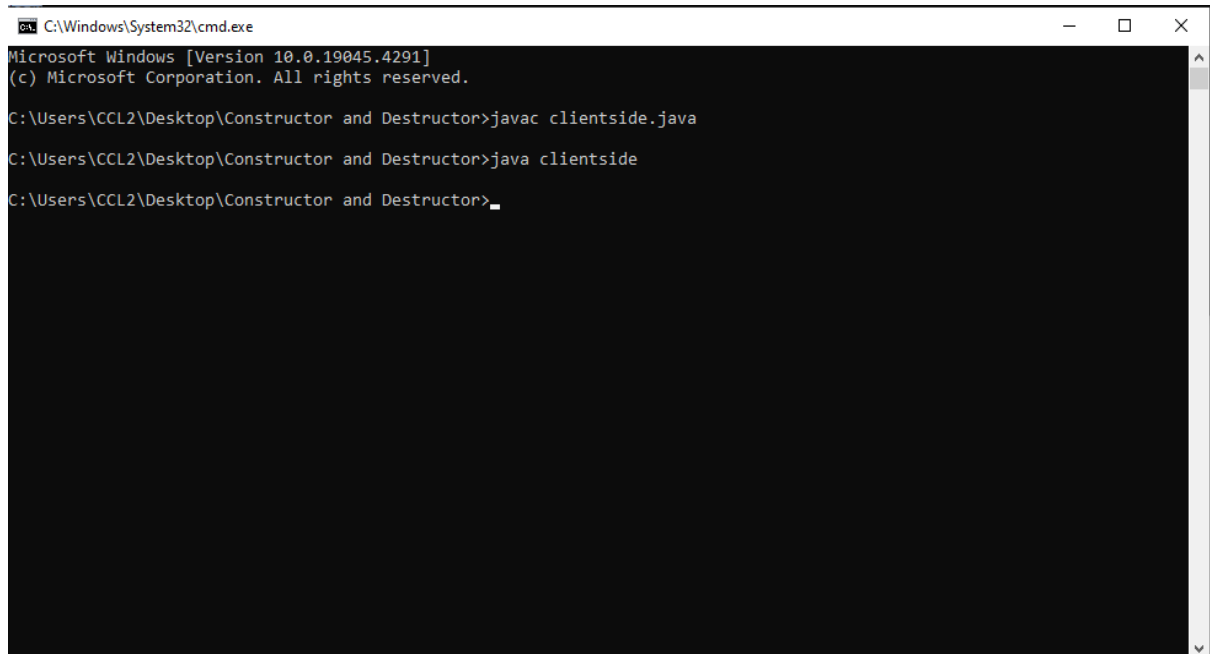
public class serverside
{
    public static void main(String[] args)
    {
        try
        {
            int port=Integer.parseInt("4444");
            DatagramSocket ds=new DatagramSocket(port);
            byte buffer[]=new byte[20];
            while(true)
            {
                DatagramPacket dp=new DatagramPacket(buffer,buffer.length);
                ds.receive(dp);
                String str=new String(dp.getData());
                System.out.println(str);
            }
        }
        catch(Exception e)
        {
            e.printStackTrace();
        }
    }
}
```

```
    }  
}  
}
```

clientside.java:

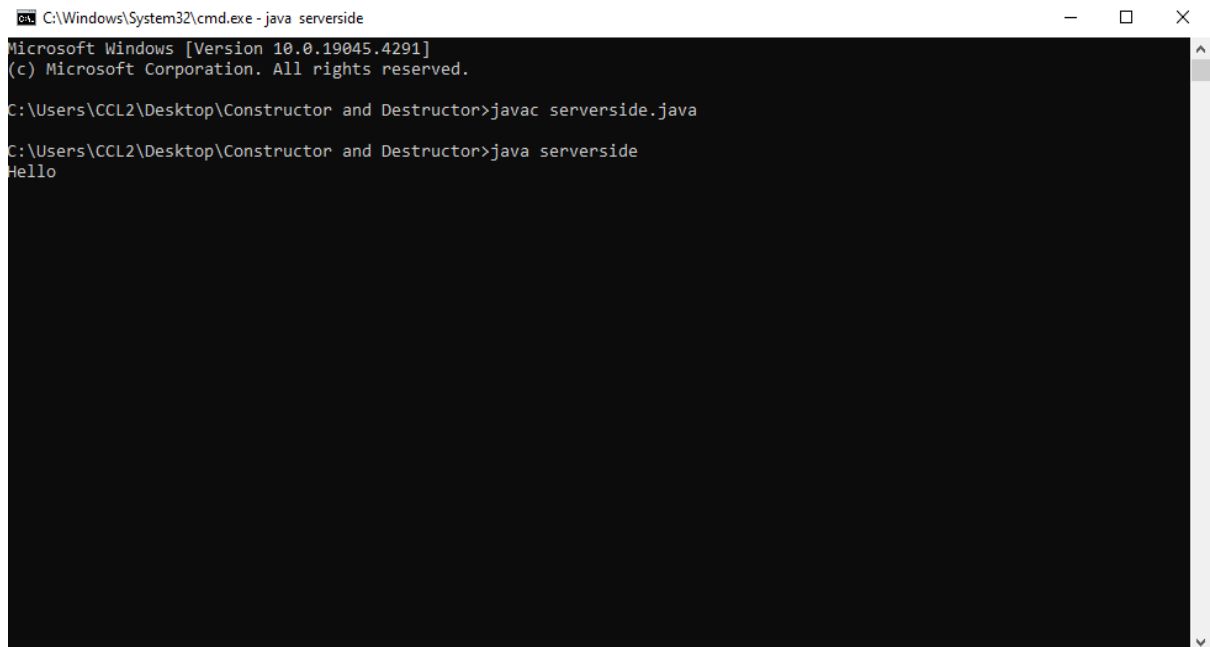
```
package sockets;  
import java.net.*;  
class clientside  
{  
public static void main(String[]args)  
{  
    try  
    {  
        InetAddress ia=InetAddress.getByName("192.168.5.122");  
        int port=Integer.parseInt("4444");  
        DatagramSocket ds=new DatagramSocket();  
        String s1="Hello";  
        byte buffer[]=s1.getBytes();  
        DatagramPacket dp=new DatagramPacket(buffer,buffer.length,ia,port);  
        ds.send(dp);  
  
    }  
    catch(Exception e)  
    {  
        System.out.println("Exception:"+e);  
    }  
}  
}
```


OUTPUT:



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CCL2\Desktop\Constructor and Destructor>javac clientside.java
C:\Users\CCL2\Desktop\Constructor and Destructor>java clientside
C:\Users\CCL2\Desktop\Constructor and Destructor>_
```



```
C:\Windows\System32\cmd.exe - java serverside
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CCL2\Desktop\Constructor and Destructor>javac serverside.java
C:\Users\CCL2\Desktop\Constructor and Destructor>java serverside
Hello
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

ABSTRACT CLASS

SOURCE CODE:

```
import java.util.*;

abstract class Alpha{

    int a,b;

    void getdata()

    {

        Scanner scn=new Scanner(System.in);

        System.out.println("Enter A value is:");

        a=scn.nextInt();

        System.out.println("Enter B value is:");

        b=scn.nextInt();

    }

    abstract void display();

}

public class Abstr extends Alpha{

    int add, sub,mul;

    void display()

    {

        add=a+b;

        sub=a-b;
```

```
mul=a*b; System.out.println("Addition value is:" +add);
```

```
System.out.println("Subtracted value is:" +sub);
```

```
System.out.println("Multiplied value is:" +mul);
```

```
}
```

```
public static void main(String[] args)
```

```
{
```

```
Abstr obj=new Abstr();
```

```
obj.getdata();
```

```
obj.display();
```

```
}
```

```
}
```

OUTPUT:

```
F:\year II>javac Abstr.java
F:\year II>java Abstr
Enter A value is:
14
Enter B value is:
24
Addition value is:38
Subtracted value is:-10
Multiplied value is:336
F:\year II>
```

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

REMOTE METHOD INVOCATION

SOURCE CODE:

RMI INTERFACE:

```
import java.rmi.*;

public interface AddServerInterface extends Remote
{double add(double d1,double d2)throws RemoteException;
}
```

RMI IMPLEMENTING CLASS:

```
import java.rmi.*;
import java.rmi.server.*;

public class AddServerImplementingClass extends UnicastRemoteObject
implements AddServerInterface{

    public AddServerImplementingClass()throws RemoteException{

    }

    public double add(double d1,double d2)throws RemoteException{

        return d1+d2;

    }

}
```

RMI SERVER SIDE:

```
import java.net.*;
import java.rmi.*;
```

```

public class AddServer
{
    public static void main(String[] args)throws RemoteException
    {
        try {
            AddServerImplementingClass obj=new AddServerImplementingClass();
            Naming.rebind("AddServer",obj);
        }
        catch(Exception e)
        {
            System.out.println("Exception"+e);
        }
    }
}

```

RMI CLIENT SIDE:

```

import java.rmi.*;
public class AddClient{
    public static void main(String[] args){
        try {
            String addServerURL="rmi://" +args[0]+"/AddServer";
            AddServerInterface
            obj=(AddServerInterface)Naming.lookup(addServerURL);
            System.out.println("the first number is="+args[1]);
            double d1=Double.parseDouble(args[1]);
            System.out.println("the second number is="+args[2]);
            double d2=Double.parseDouble(args[2]);
            System.out.println("the sum is"+obj.add(d1,d2));
        }
    }
}

```

```
}  
catch(Exception e)  
{  
    System.out.println("Exception="+e);  
}  
}  
}
```

OUTPUT:

```
C:\Windows\System32\cmd.exe - java AddServer
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

F:\raj>set path=C:\Program Files (x86)\Java\jdk1.7.0\bin
F:\raj>javac AddServerInterface.java
F:\raj>javac AddServerImplementingClass.java
F:\raj>javac AddServer.java
F:\raj>javac AddClient.java
F:\raj>rmic AddServerImplementingClass
F:\raj>start rmiregistry
F:\raj>java AddServer
```



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

F:\raj>set path=C:\Program Files (x86)\Java\jdk1.7.0\bin
F:\raj>java AddClient 127.0.0.1 14 24
the first number is=14
the second number is=24
the sum is38.0
F:\raj>
```


REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

APPLET

SOURCE CODE:

```
package applet;

import java.awt.*;
import java.awt.event.*;
import java.applet.*;
import java.awt.color.*;

public class signal extends Applet implements ActionListener
{
    int colorNum;

    Button btn1=new Button ("stop traffic");
    Button btn2=new Button ("caution");
    Button btn3=new Button ("proceed");
    // Button btn4=new Button ("stop traffic");
    // Button btn5=new Button ("caution");
    // Button btn6=new Button ("proceed");

    public void init()
    {
        setBackground(Color.LIGHT_GRAY);
        btn1.addActionListener(this);
        btn2.addActionListener(this);
        btn3.addActionListener(this);
        add(btn1);
        add(btn2);
```

```
        add(btn3);

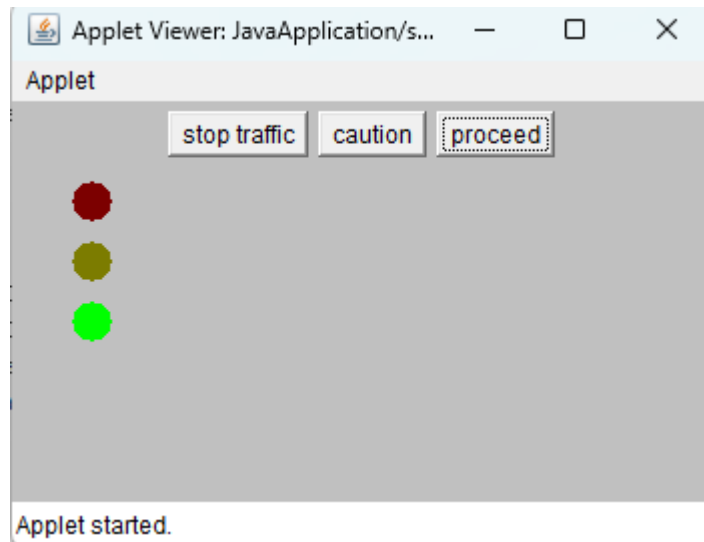
    }

    public void paint(Graphics g)
    {
        g.setColor(Color.BLACK);
        g.setColor(colorNum==1?Color.red:Color.red.darker().darker());
        g.fillOval(30,40,20,20);
        g.setColor(colorNum==2?Color.yellow:Color.yellow.darker().darker());
        g.fillOval(30,70,20,20);
        g.setColor(colorNum==3?Color.green:Color.green.darker().darker());
        g.fillOval(30,100,20,20);

    }

    public void actionPerformed(ActionEvent evt)
    {
        if(evt.getSource()==btn1)
            colorNum=1;
        else if(evt.getSource()==btn2)
            colorNum=2;
        else
            colorNum=3;
        repaint();
    }
}
```

OUTPUT:



REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

SWING

SOURCE CODE:

```
package swinglogin;

import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JPasswordField;
import javax.swing.JTextField;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class Swinglogin {
    public static void main(String[] args)
    {
        JFrame frame=new JFrame("Login Screen"); //Line 1
        JLabel label = new JLabel();//Line 2
        final JLabel tf= new JLabel();

        label.setBounds(20,150, 200,50);

        //Line 3

        final JPasswordField value = new JPasswordField();//Line 4
        value.setBounds(100,75,100,30);
```

```

JLabel userNameLabel=new JLabel("User Name:");//Line 5
userNameLabel.setBounds(20,20, 80,30);

JLabel paswordLabel=new JLabel("Password:");//Line 6
paswordLabel.setBounds(20,75, 80,30);

JButton loginButton = new JButton("Login Here");//Line 7
loginButton.setBounds(100,120, 80,30);

final JTextField text = new JTextField();//Line 8
text.setBounds(100,20, 100,30);
tf.setBounds(200,120,500,130);

frame.add(value); //Line 9
frame.add(userNameLabel);
frame.add(label); //Line 9
frame.add(paswordLabel); //Line 9
frame.add(loginButton); //Line 9
frame.add(text);
frame.add(tf);//Line 9
frame.setSize(400,400);
frame.setLayout(null); //Line 10
frame.setVisible(true); //Line

loginButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String userNameDisplay;
        userNameDisplay = text.getText()+" Login are Created";
        tf.setText(userNameDisplay);
//System.out.println(userNameDisplay);
    }
});

}}

```

OUTPUT:

The image shows a Java Swing window titled "Login Screen". Inside the window, there is a login form. The "User Name:" label is followed by a text input field containing the text "Siraj". The "Password:" label is followed by a password input field represented by a series of dots. Below the password field is a button labeled "Login ...". At the bottom right of the window, the text "Siraj Login are Created" is displayed.

REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

CALCULATOR

SOURCE CODE:

```
package Calculator;

import java.awt.event.*;
import javax.swing.*;
import java.awt.*;

class Calculator extends JFrame implements ActionListener
{
    static JFrame f;
    static JTextField l;
    String s0, s1, s2;
    Calculator()
    {
        s0 = s1 = s2 = " ";
    }
    public static void main(String args[])
    {
        f = new JFrame("Calculator");
        try
        {
            UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());
        }
        catch (Exception e)
        {
            System.err.println(e.getMessage());
        }
    }
}
```

```
}  
  
Calculator C = new Calculator();  
  
l = new JTextField(16);  
  
l.setEditable(false);  
  
JButton b0, b1, b2, b3, b4, b5, b6, b7, b8, b9, ba, bs, bd, bm, be, beq, beq1;  
  
b0 = new JButton("0");  
  
b1 = new JButton("1");  
  
b2 = new JButton("2");  
  
b3 = new JButton("3");  
  
b4 = new JButton("4");  
  
b5 = new JButton("5");  
  
b6 = new JButton("6");  
  
b7 = new JButton("7");  
  
b8 = new JButton("8");  
  
b9 = new JButton("9");  
  
beq1 = new JButton("=");  
  
ba = new JButton("+");  
  
bs = new JButton("-");  
  
bd = new JButton("/");  
  
bm = new JButton("*");  
  
beq = new JButton("C");  
  
be = new JButton(".");  
  
JPanel p = new JPanel();  
  
bm.addActionListener(C);  
  
bd.addActionListener(C);  
  
bs.addActionListener(C);  
  
ba.addActionListener(C);  
  
b9.addActionListener(C);
```



```
b8.addActionListener(C);
b7.addActionListener(C);
b6.addActionListener(C);
b5.addActionListener(C);
b4.addActionListener(C);
b3.addActionListener(C);
b2.addActionListener(C);
b1.addActionListener(C);
b0.addActionListener(C);
be.addActionListener(C);
beq.addActionListener(C);
beq1.addActionListener(C);
p.add(l);
p.add(ba);
p.add(b1);
p.add(b2);
p.add(b3);
p.add(bs);
p.add(b4);
p.add(b5);
p.add(b6);
p.add(bm);
p.add(b7);
p.add(b8);
p.add(b9);
p.add(bd);
p.add(be);
p.add(b0);
```

```

p.add(beq);

p.add(beq1);

p.setBackground(Color.blue);

f.add(p);

f.setSize(200, 220);

f.show();

}

public void actionPerformed(ActionEvent e)

{

String s = e.getActionCommand();

if ((s.charAt(0) >= '0' && s.charAt(0) <= '9') || s.charAt(0) == '.')

{

if (!s1.equals(""))

s2 = s2 + s;

else

s0 = s0 + s;

l.setText(s0 + s1 + s2);

}

else if (s.charAt(0) == 'C')

{

s0 = s1 = s2 = "";

l.setText(s0 + s1 + s2);

}

else if (s.charAt(0) == '=')

{

double te;

if (s1.equals("+"))

te = (Double.parseDouble(s0) + Double.parseDouble(s2));

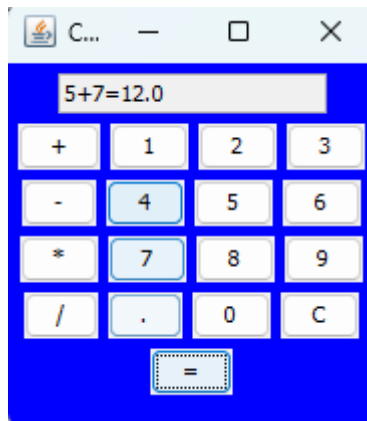
```

```

else if (s1.equals("-"))
    te = (Double.parseDouble(s0) - Double.parseDouble(s2));
else if (s1.equals("/"))
    te = (Double.parseDouble(s0) / Double.parseDouble(s2));
else
    te = (Double.parseDouble(s0) * Double.parseDouble(s2));
l.setText(s0 + s1 + s2 + "=" + te);
s0 = Double.toString(te);
s1 = s2 = "";
}
else {
    if (s1.equals("") || s2.equals(""))
        s1 = s;
    else {
        double te;
        if (s1.equals("+"))
            te = (Double.parseDouble(s0) + Double.parseDouble(s2));
        else if (s1.equals("-"))
            te = (Double.parseDouble(s0) - Double.parseDouble(s2));
        else if (s1.equals("/"))
            te = (Double.parseDouble(s0) / Double.parseDouble(s2));
        te = (Double.parseDouble(s0) * Double.parseDouble(s2));
        s0 = Double.toString(te);
        s1 = s;
        s2 = "";
    }
    l.setText(s0 + s1 + s2);
}
}}

```

OUTPUT:



REG. NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN
YEAR/SEM/SEC: Y2/S4/B

JDBC

SOURCE CODE:

```
package jdbcempdetails;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.ResultSet;

import java.sql.Statement;

public class JDBCempDetails

{

    public static void main(String args[])

    {

        try

        {

            Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");

            Connection con =

DriverManager.getConnection("jdbc:ucanaccess://empdetail.accdb");

            System.out.println(con);

            Statement s = con.createStatement();

            String query = "select * from Employee";

            s.execute(query);

            ResultSet rs = s.getResultSet();

            while(rs.next())

            {

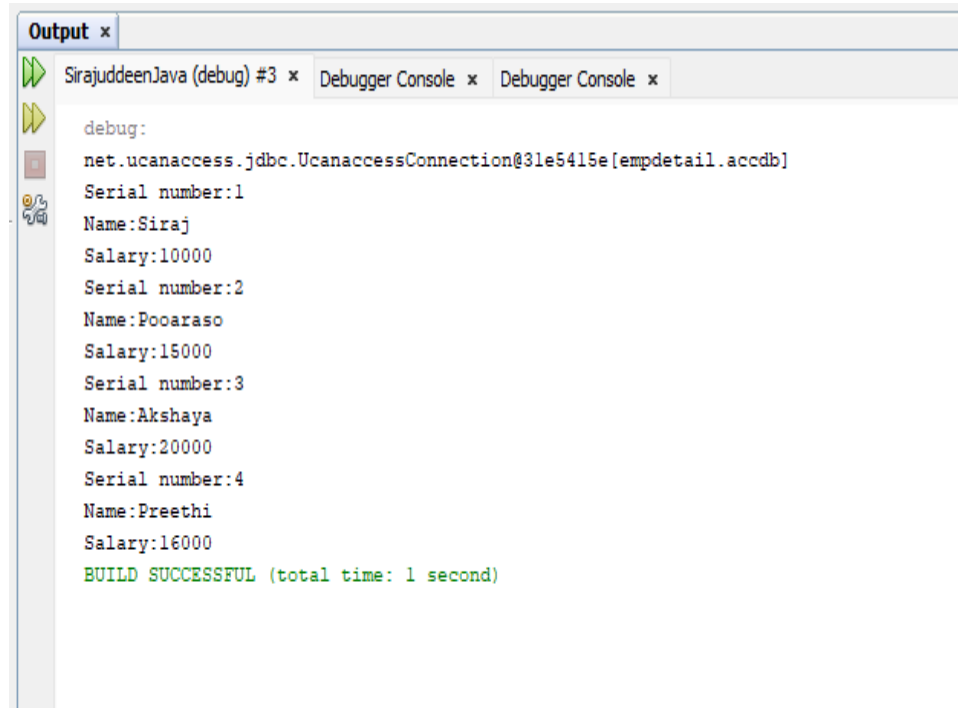
                System.out.println("Serial number:"

+rs.getString(1));

                System.out.println("Name:"+rs.getString(2));
```

```
        System.out.println("Salary:"  
        +rs.getString(3));  
  
    }  
    s.close();  
    con.close();  
  
}  
catch(Exception e)  
{  
    System.out.println("exception"+e);  
}  
}  
}
```

OUTPUT:



```
Output x
SirajuddeenJava (debug) #3 x  Debugger Console x  Debugger Console x

debug:
net.ucanaccess.jdbc.UcanaccessConnection@31e5415e[empdetail.accdb]
Serial number:1
Name:Siraj
Salary:10000
Serial number:2
Name:Pooaraso
Salary:15000
Serial number:3
Name:Akshaya
Salary:20000
Serial number:4
Name:Preethi
Salary:16000
BUILD SUCCESSFUL (total time: 1 second)
```